KNOWLEDGE CHECK

1) What is a data type in the context of programming and operating systems?

It tells the type of value a variable holds (like number or letter) and how much memory it needs.

2) How many bytes does the OS allocate for a char data type? How many bytes for a double?

char = 1 byte, double = 8 bytes.

3) Name the five main memory segments in a process memory layout.

Text, Data, BSS, Heap, Stack.

4) What is the difference between the Data segment and the BSS segment?

Data stores variables with a starting value; BSS stores variables with no starting value (set to 0).

5) Why does the OS place the stack at high virtual addresses and the text segment at low addresses?

The text segment is allocated at the lowest virtual addresses because it is immutable (fixed in size). The stack segment is allocated at the highest virtual addresses because it is dynamic

6) What happens to stack memory when a function returns? Who reclaims it?

It is automatically freed by the OS.

7) What is the risk of forgetting to call free() on heap memory?

It causes a memory leak (wasted memory).

8) Explain the difference between ptr and &ptr when printing addresses.

ptr is the address it points to; &ptr is the address of the pointer itself.

9) Why is the cast (void *) used when printing pointers with %p?

%p only reads void * pointers

10) How does the OS know which memory belongs to which process?

Each process has its own memory space, tracked by the OS using page tables.